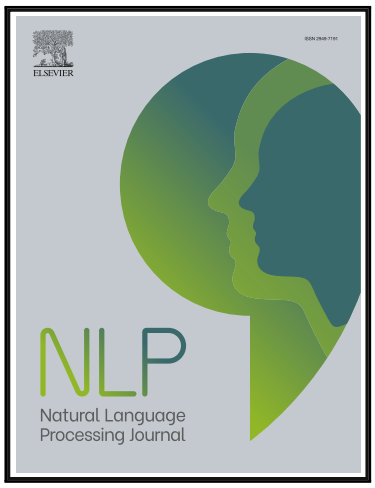
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PII: S2949-7191(26)00015-4
DOI: https://doi.org/10.1016/j.nlp.2026.100210
Reference: NLP 100210

To appear in: *Natural Language Processing Journal*

Received date: 3 September 2025
Revised date: 4 April 2026
Accepted date: 25 April 2026

Please cite this article as: Jinyu Wang , Zenren Yu , Haodong Li , Hongjun Lin , Jing Li , The STMO Model for Balancing Growth and Conservation in Juneau, Alaska, *Natural Language Processing Journal* (2026), doi: https://doi.org/10.1016/j.nlp.2026.100210

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The STMO Model for Balancing Growth and Conservation in Juneau, Alaska

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#The authors make equal contributions to the study

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Competing Interests

The investigators claim no competing interests.

Authorship Contribution Statement

Jing Li: Writing-Original draft preparation, Conceptualization, Supervision, Project administration.

Jinyu Wang: Methodology.

Zenren Yu: Validation.

Haodong Li: Software.

Hongjun Lin: Language review.

Data Availability

On request.

Declarations

Inapplicable.

Conflicts of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Author Statement

All investigators have examined and authorized the manuscript, fulfilling the criteria for authorship.

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Each investigator certifies that it reflects honest work.

Funding

This study was supported by China College Students Innovation and Entrepreneurship Program (No. 202412617003X) and Application of continuous frameworks in the principle of uncertainty (No. 2023A04J1328)

Ethical Approval

All investigators have substantially contributed to the important work involved in the article and are willing to assume public accountability for its content.

Abstract

This paper offers a dynamic model to handle the issues of sustainable tourism management in Juneau, Alaska, a city dealing with overtourism. The model minimizes negative effects by combining social, environmental, and financial aspects to maximize tourism income. Dynamic feedback loops and a multi-objective optimization framework help to balance visitor counts, income, and sustainability. The first section of the research finds ideal visitor numbers and tax rates to maximize income while reducing social strain and environmental damage. The second section looks at how reinvesting tourism money into infrastructure and sustainability initiatives might help to reduce traffic congestion, improve environmental quality, and encourage local acceptance. The third component does a sensitivity study to find important factors that greatly affect sustainable results, such as environmental mitigating action and infrastructure efficiency. The model is meant to be flexible, which helps it to be used at other popular tourist sites. It describes plans to distribute visitor flows, boost lesser-known attractions, and make calculated investments to enhance the whole travel experience. Applying this strategy guarantees long-term sustainability of tourism by arming legislators with useful tools to balance economic gains with environmental protection and community well-being, preserving cultural and natural assets.

Keywords: sustainable tourism, overtourism, Juneau, multi-objective optimization, dynamic feedback, sensitivity analysis, reinvestment strategies, tourism management

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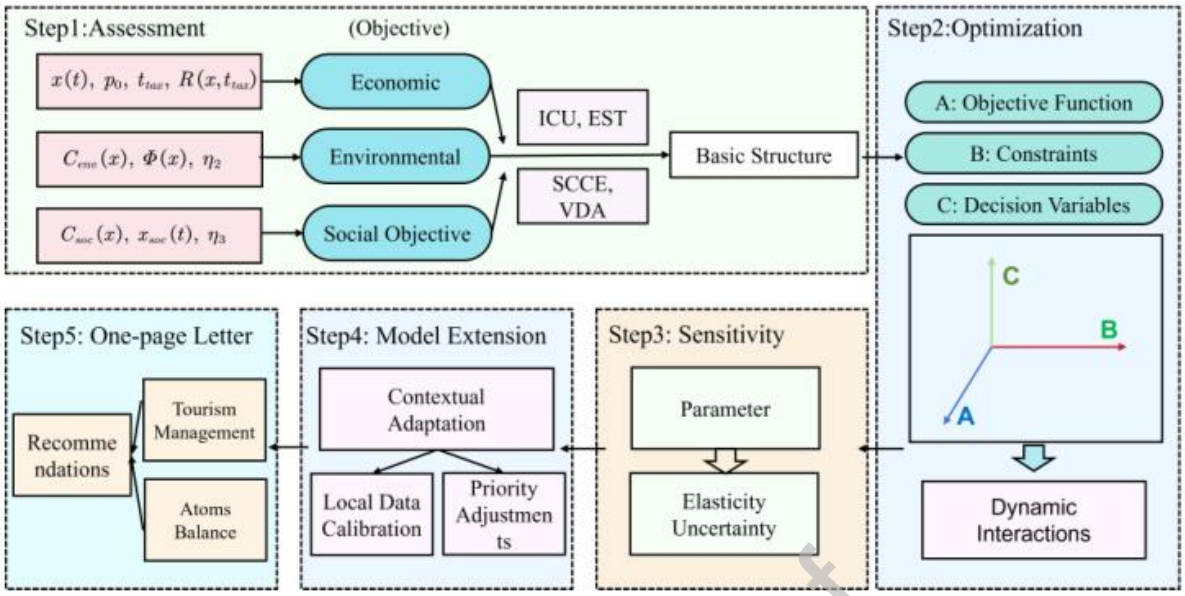


Fig. 1. Framework of sustainable tourism management and optimization model in this paper

1. Introduction

Juneau, Alaska, had a record-breaking 1.6 million cruise visitors in 2023, earning $375 million in income with a population of roughly 30,000. But this flood of visitors results in congestion and environmental problems, including the shrinking Mendenhall Glacier, partially brought on by overtourism. Although Juneau has other attractions, including rainforests and whale viewing, sustainable tourism is vital for the city's future. The first has to take care of hidden tourism costs, including carbon footprints, waste management, and infrastructure strain. Measures including hotel taxes, visitor fees, and visitor caps have been taken to help lighten the load; yet, some residents have expressed worries about the effect of extra fees. Hidden costs have to be included in a model for sustainable tourism in Juneau, and a balanced approach to tourism development has to be supported while variables, including visitor numbers, revenue, and stabilizing actions, are considered.

1.1. Literature Review

Many places have benefited economically from the increase in world travel, but it has also presented major problems, especially in places suffering from overtourism. Such a place is Juneau, Alaska, which struggles with infrastructure strain, environmental damage, and social conflict brought on by visitor count [1]. Zuo et al. [2] studied the problem of overtourism in Juneau, which faced high pressures from 1.6 million cruise ship arrivals in 2023, bringing in $375 million in revenue but straining the infrastructure, environment, and society. In order to

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alleviate this set of issues, the researchers proposed a multi-objective optimization method to promote sustainable tourism development through management of tourist quantities, tax rates, and implementation of measures for sustainable environments. The proposed mathematical model is based on decision variables like the amount of tourist arrivals and the tax ratio, subject to constraints on infrastructure, environments, and social satisfaction. The Hippo optimization technique is used to determine optimal values, and the outcomes show that by restricting the entry of tourists to no more than 9,701 and increasing the tax rate to 17.9%, the objectives can be satisfied. The proposed mathematical model is later used on the city of Venice, introducing an inequality constraint on housing prices, and the optimization outcome shows that by restricting the entry amount to no more than 54,100 and increasing the tax rate to 7.9%, infrastructure loss, degradation of environments, and residential satisfaction, as well as housing prices, are all optimized.

Researchers have created several models combining social, environmental, and financial factors to meet these difficulties. Using stationary optimization, dynamic programming, and the Analytic Hierarchy Process (AHP), Rosardi et al. [3] put out a thorough plan for sustainable travel management in Juneau. Their approach suggests ideal visitor counts and investment in environmental and social sustainability by balancing economic development with these factors.

The STMO model, while integrating social, environmental, and financial objectives through multi-objective optimization and feedback systems, can be strengthened by engaging with established theoretical frameworks in tourism, planning, and sustainability. For instance, Butler's Tourism Area Life Cycle [4] provides valuable insights into the evolving stages of tourism destinations, which can be used to tailor optimization strategies based on Juneau's tourism development phase. Additionally, Dwyer and Kim's Competitiveness Model [5] emphasizes the interplay of economic, social, and environmental factors in ensuring long-term destination success, offering a way to balance sustainability with competitiveness. Furthermore, integrating stakeholder perception approaches [6], [7] highlights the importance of considering the views of local communities and tourists in shaping policies that are socially acceptable and environmentally effective. By incorporating these frameworks, the STMO model can better address the complexities of sustainable tourism governance, ensuring that optimization strategies are both theoretically grounded and practically applicable to real-world tourism

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management.

Pan et al. [8] proposed a multi-dimensional sustainable tourism optimization model for Juneau, Alaska, focusing on striking a balance between the economic prosperity generated by tourism and sustainability. The proposed model uses a hybrid Analytic Hierarchy Process and Entropy-weight approach, where the optimized sustainable tourism model uses the NSGA-II algorithm to achieve an optimal tourist limit through tax variation, keeping the economic revenue constant at USD 325 million and channeling it back into sustainable conservation and social infrastructure. The proposed optimization model combats both the degradation of the environment, including glacier retreat, and the associated issues of infrastructure and citizen grievances against tourism. The applicability of this optimization model was verified in the case of Barcelona, Spain, where the problem of urban overtourism was reduced by decreasing housing and traffic expenses, along with improved environmental sustainability. The sensitivity tests verified the sustainability and robustness of this optimization model, and future scopes are identified by integrating real-time analytics techniques and dynamic modeling approaches. The crucial policy implications and recommendations are provided by focusing on adaptive tourist capacity, attraction diversification, and citizen engagement platforms, which are feasible and scalable solutions for destinations all over the globe.

Arbolino et al. [9] carried out their research on strategic management within the tourism industry and underlined the value of appropriate planning for the sustainable implementation of tourism activities, given the complexity of the decision-making process and various interested parties. The authors present the applicability of the proposed Multi-objective Optimization decision model on projects proposed for funding within public administrations, with the scope of maximizing the efficiency of public resource utilization. The proposed decision model takes into account various criteria, including environmental, social, and economic, in order to select those tourism activities that maximize the interests' parties' utility. The proposed decision model was validated through the case study of a tender prepared by an Italian public administration to improve the attractiveness of urban areas. The outcome of this research illustrated the efficiency of the proposed decision model compared with other decision models used within multicriteria decision analysis, as it achieves both quantitatively and qualitatively superior allocation of public funds. Corboș et al. [10] have described the full

significance of investments in tourism infrastructure and their overall contribution to the growth and development of the tourism sector through an extensive review of literature. The authors strongly underline the overall significance of infrastructure investments, ranging from transportation, accommodation, and sports infrastructure, and their contribution to retaining and maintaining the attraction of tourists. The authors mention the overall implications of these investments on the overall experience of the tourist as well as on the economy of the place. The authors also enumerate issues and constraints emerging from the implementation of infrastructure investments, ranging from sustainability and overall balance between infrastructure development and conservation of natural resources through extensive reviews and discussions of various theoretical and qualitative studies and research in this sector.

Comparably, Pitakaso et al. [11] investigated sustainable tourism using a mathematical model combining ESG criteria, Sustainable Development Goals (SDGs), and the Global Reporting Initiative (GRI) system. Their research highlights the need for legislative instruments, including taxes and caps, in reducing environmental effects while preserving economic gains. Planning sustainable tourism has also benefited from multi-objective optimization techniques. Published in the Journal of Cleaner Production, a paper presented a fresh multi-objective optimization method for creating customized travel itineraries balancing costs, visitor preferences, and environmental aspects [12]. In the framework of overtourism, the World Tourism Organization (UNWTO) emphasizes tactics including spreading visitors over attractions and time, supporting fresh itineraries, and including locals in travel policy decisions. These plans seek to lessen the negative effects of overtourism and promote environmentally friendly travel [13].

The STMO model, while integrating social, environmental, and financial objectives through multi-objective optimization and feedback systems, can be strengthened by engaging with established theoretical frameworks in tourism, planning, and sustainability. For instance, Butler's Tourism Area Life Cycle (1980) provides valuable insights into the evolving stages of tourism destinations, which can be used to tailor optimization strategies based on Juneau’s tourism development phase. Additionally, Dwyer and Kim's Competitiveness Model (2003) emphasizes the interplay of economic, social, and environmental factors in ensuring long-term destination success, offering a way to balance sustainability with competitiveness. Furthermore,

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integrating stakeholder perception approaches (Nag & Mishra, 2023; 2024) highlights the importance of considering the views of local communities and tourists in shaping policies that are socially acceptable and environmentally effective. By incorporating these frameworks, the STMO model can better address the complexities of sustainable tourism governance, ensuring that optimization strategies are both theoretically grounded and practically applicable to real-world tourism management.

Sustainable tourism management in places like Juneau, impacted by overtourism, calls for integrated models weighing social, environmental, and financial elements. Achieving a balance between tourism development and sustainability depends on using multi-objective optimization models and including local stakeholders.

1.2. Research gaps

Several research gaps remain in sustainable tourism management. Many models are theoretical or simulation-based, lacking real-world validation and cross-city comparisons. There is a need for more integrated approaches that combine economic, social, and environmental factors, as well as research into tourist behavior, demand adaptation, and the use of real-time data in dynamic models. The long-term effects of reinvestment strategies, especially on community well-being and environmental sustainability, require further exploration. Additionally, the impact of tourism on housing markets and social equity is often overlooked. Research should also focus on adapting policies to different cultural and regional contexts, incorporating stakeholder involvement, and balancing economic profitability with tourist satisfaction. Finally, optimizing models for scalability across diverse global destinations is essential.

1.3. Contribution and Innovation

This study integrates social, environmental, and financial sustainability into a single tourism management model, offering a dynamic, multi-objective optimization framework for controlling overtourism in Juneau, Alaska. Unlike previous models that sometimes concentrated on static or single-objective approaches, this study provides a temporally adaptive framework that develops with feedback mechanisms, allowing legislators to track and react to changing tourism impacts in real time. Furthermore, offering a clear road map for sustainable

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reinvestment planning, the model quantitatively links reinvestment strategies, such as infrastructure improvements, environmental restoration, and social initiatives, to long-term increases in capacity, ecosystem health, and community tolerance. The innovative feature of this work is its integration of dynamic feedback systems inside a multi-objective tourism management model, facilitating adaptive policy simulation over several years. The model specifically allows a granular study of reinvestment results by introducing separate efficiency parameters (α, β, γ) for infrastructure, environmental health, and social carrying capacity to impact each. Moreover, it can especially replicate behavioral reactions to tourism policies since it comprises a demand adaptation mechanism that changes visitor flow depending on environmental quality, social conditions, and pricing. Existing tourism models rarely reflect this degree of adaptability and accuracy.

2. Model Assumptions and Key Variables

2.1.Basic Assumptions

1. Tourist demand is both price- and environment-sensitive. Rising taxes or fees for visits can be reduced, while declining environmental quality, that is, declining glaciers and pollution, can discourage future visitors. The elasticity of demand concerning price and surroundings will affect the number of visitors.

2. Local infrastructure has limited capacity. There exists a threshold (e.g., maximum daily or annual **visitors**) beyond which additional costs and negative impacts on local facilities (e.g., waste management, water supply) escalate significantly, leading to congestion and reduced service quality.

3. Environmental degradation is linked to carbon emissions and other forms of pollution. Emissions and pollution produced by cruise ships, flights, and other tourist activities can build up over time and have long-term effects on the environment, including habitat disturbance, resource depletion, and worsening air quality.

4. Social carrying capacity is finite. Once this threshold is exceeded, discontent, local protests, and possible social unrest may result, thus lowering the general social welfare and the quality of life for the residents. Residents can only tolerate a certain volume of tourists and related behaviors (e.g., overcrowding, noise).

5. Reinvestment in sustainability measures can mitigate negative impacts. A fraction of

the tourism income, that from taxes or fees, should go toward sustainability initiatives such as projects for community development, infrastructure enhancements, and environmental protection. These expenditures can support long-term social acceptance and environmental health, as well as help to lessen the bad effects of tourism.

2.2.Key Variables and Parameters

In this section, the parameters and numbers will be examined.

Table 1. Key variables and parameters

① Core Variable, ② Cost/Capacity Variable, ③ Dynamic Feedback Parameter, ④ Elasticity Parameter.

Type	Symbol	Description
①	$x(t)$	Number of visitors (or cruise ship arrivals) in period t. This is a key driver for both revenue generation and environmental/social impacts.
①	$R(x, t_{\text{tax}})$	Total tourism revenue in period t, calculated as $[p_0 + t_{\text{tax}}] \cdot x(t)$, where p_0 is the baseline per-visitor spending (in dollars) and t_{tax} is the per-visitor tax or fee. This represents the economic revenue generated from tourism.
①	t_{tax}	Per-visitor tax or fee (e.g., environmental levy, head tax). The value of t_{tax} will influence the visitor demand and tourism-related revenue.
①	p_0	Baseline per-visitor spending, excluding taxes or fees. This reflects the average spending of a tourist, primarily on accommodation, food, and activities.
②	$C_{\text{env}}(t)$	Environmental cost in period t, representing the cost due to pollution, carbon emissions, resource consumption, etc., which generally increases with the number of visitors, $x(t)$.
②	$C_{\text{cos}}(t)$	Social cost in period t, capturing negative impacts like overcrowding, increases in housing prices, and general resident dissatisfaction. This parameter reflects the social tolerance or dissatisfaction with tourism.
②	$I_{\text{cap}}(t)$	Infrastructure capacity at time t, representing the maximum number of visitors that local facilities (e.g., waste management, water supply, public transport) can handle without significant strain.
②	$x_{\text{soc_max}}(t)$	Social carrying capacity at time t, denoting the maximum number of visitors that residents are willing to tolerate, beyond which social unrest or dissatisfaction may arise.
③	δ	Fraction of tourism-generated revenue reinvested in sustainability measures (e.g., infrastructure upgrades, environmental restoration, community

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		programs). This is a key parameter for managing the reinvestment of revenue to mitigate tourism's negative effects.
③	$\Phi(x)$	Environmental degradation function caused by $x(t)$ visitors in period t. This function represents the accumulated negative environmental impacts, such as emissions, resource depletion, or habitat destruction, due to tourism activities.
③	η_1	Efficiency of reinvestments in increasing infrastructure capacity (e.g., funds-to-capacity conversion rate). A higher η_1 suggests that the reinvested funds lead to more efficient infrastructure growth, increasing the capacity to handle more tourists.
③	η_2	Efficiency of reinvestments in mitigating environmental costs (e.g., funds-to-environmental-repair efficiency). This parameter measures the effectiveness of funds in reducing environmental degradation, such as improving waste treatment or restoring ecosystems.
③	η_3	Efficiency of reinvestments in improving social carrying capacity (e.g., funds-to-community-benefit efficiency). This parameter represents the effectiveness of reinvestments in improving social tolerance, including reducing crowding and housing pressures, which can enhance residents' acceptance of tourism.
④	α	Price elasticity of visitor demand, representing how sensitive tourist arrivals are to changes in the per-visitor tax t_{tax}. A higher α means that visitors are more responsive to changes in price/tax, which may be used to regulate demand.
④	β	Environmental elasticity of visitor demand, capturing how sensitive tourist arrivals are to changes in environmental quality $C_{\text{env}}(t)$. This indicates how tourism demand changes in response to environmental degradation (e.g., receding glaciers, pollution).
④	γ	Social elasticity of visitor demand, reflecting how sensitive tourist arrivals are to changes in social conditions $C_{\text{soc}}(t)$. This parameter represents how visitors adjust their behavior based on crowding, social unrest, or local satisfaction with tourism.
①	$S(t)$	State vector at time u, including infrastructure capacity $I_{\text{cap}}(t)$, environmental cost $C_{\text{env}}(t)$, and social carrying capacity $x_{\text{soc_max}}(t)$. This vector represents the current state of the system and is critical for understanding the sustainability of tourism in the area.
①	$\boldsymbol{u}(t)$	Control vector at time u, including the number of visitors $x(t)$, pervisitor

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tax $t_{\text{tax}}(t)$, and reinvestment fraction $\delta(t)$. These are the control variables that can be adjusted to optimize the sustainability model.

3. Model Structure and Core Equations

3.1.Multi-Objective Optimization Framework

Following the clarification of the research background, basic assumptions, and important variables, a whole model was constructed combining social, environmental, and financial goals using a multi-objective optimization technique. This model captures over time the dynamic interactions among tourism development, environmental elements, and social aspects. The model is developed not only to maximize the total benefits in a single period (or season) but also to include dynamic feedback equations showing the evolution of infrastructure capacity, environmental quality, and social carrying capacity over several periods, so accounting for the long-term consequences of tourism policies.

This framework builds on O'Reilly's [14] Tourism Carrying Capacity framework for infrastructure and social constraints, as well as Forrester's [15] systems dynamics model to simulate the long-term interactions between tourism policies and resource management.

3.1.1. Decision Variables

The following decision variables are essential to the model:

- $x(t)$: The number of visitors (or cruise ship arrivals) permitted or realized in period t.
- t_{tax}: The per-visitor fee or tax (e.g., an environmental levy or head tax).
- δ: The fraction of tourism-generated tax revenue (ranging from 0 to 1) reinvested in sustainability measures, such as infrastructure upgrades, environmental protection, and community welfare.

3.1.2. Objective Functions

The following defines the functions to handle the social, environmental, and financial goals: - Economic Objective: The total tourism revenue $R(x, t_{\text{tax}})$ is given by:

$$R(x, t_{\text{tax}}) = (p_0 + t_{\text{tax}}) \cdot x \quad (1)$$

where p_0 is the baseline per-visitor spending, excluding taxes, and x is the number of visitors.

Environmental Objective: The environmental cost, $C_{\text{env}}(x)$, which accounts for pollution,

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carbon emissions, and resource consumption, increases with the number of visitors:

$$C_{env}(x) = a_1 x + a_2 x^2 \quad (2)$$

This cost typically grows non-linearly with y due to increasing environmental degradation.

- Social Objective: The social cost, $C_{\text{soc}}(x)$, includes factors such as overcrowding, noise, and rising housing costs, which also increase with x:

$$C_{soc}(x) = b_1 x + b_2 x^2 \quad (3)$$

3.1.3. Composite Objective Function

To simultaneously optimize the economic, environmental, and social objectives, a composite objective function is introduced. The goal is to ensure that the total net benefit is maximized while the associated costs are accounted for:

$$\max_{x,t_{tax},\delta} [w_1 R(x, t_{tax}) - w_2 C_{env}(x) - w_3 C_{soc}(x)] \quad (4)$$

where w_1, w_2, and w_3 represent the weights assigned to the economic, environmental, and social objectives, respectively. Alternatively, a Pareto-based approach can be used, in which no explicit weighting is required, and the goal is to identify a set of Pareto-optimal solutions.

3.1.4. Constraints

The model includes several constraints to ensure the feasibility and sustainability of the tourism management policies:

Infrastructure Capacity: $x(t) \le I_{\text{cap}}(t)$, (C1)

Social Carrying Capacity: $x(t) \le x_{\text{soc_max}}(t)$, (C2)

Environmental Limits: $C_{\text{env}}(x(t)) \le C_{\text{env_max}}$, (C3)

Allocation Fraction: $0 \le \delta \le 1$. (C4)

These constraints ensure that the number of visitors does not exceed the infrastructure capacity or social carrying capacity and that environmental impacts are within acceptable limits.

3.1.5. Solution Approaches

Several methods can be employed to solve or analyze this multi-objective problem:

1. **Simulation:** Set policies (e.g., t_{tax} and δ) and iterate over time to assess how visitor flow, revenue, and costs evolve under various scenarios.

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2. **Multi-Objective Optimization Algorithms:** Use methods like evolutionary algorithms (e.g., NSGA-II), particle swarm optimization, or gradient-based approaches to search for optimal or Pareto-optimal solutions.
3. **Sensitivity Analysis:** Vary key parameters (e.g., price elasticity, cost coefficients) to analyze how uncertainty in these parameters affects the optimal solutions, identifying key factors for further research.

This multi-objective model provides a structured framework for evaluating how different tourism management policies can balance short-term economic gains with long-term environmental sustainability and social welfare. It offers a quantitative foundation for informed decision-making in destinations experiencing overtourism pressures.

Additionally, this framework can be extended with dynamic feedback mechanisms, allowing the model to capture how infrastructure capacity, environmental quality, and social tolerance evolve in response to changing visitor flows and policy decisions.

3.2.Dynamic Feedback Mechanisms

A distinctive feature of sustainable tourism is reflected in the intertemporal interactions among economic outcomes, environmental conditions, and social dynamics. The stationary multi-objective formulation is enhanced with dynamic feedback systems that change over several periods (e.g., years or seasons) to record these interactions [16]. Below is a list of several important feedback systems that link social attitudes, environmental conditions, and present-period choices to future capacity.

3.2.1. Infrastructure Capacity Updates

Improving infrastructure (e.g., waste treatment systems, public transportation, or water supply) can mitigate the negative impacts of tourism in subsequent periods. Specifically, the evolution of infrastructure capacity $I_{\text{cap}}(t)$ is represented as:

$$I_{cap}(t+1) = I_{cap}(t) = \eta_1 \cdot \delta \cdot R(x(t), t_{tax}) \quad (5)$$

where $\delta \cdot R$ is the portion of the tourism-generated revenue reinvested in capacity upgrades, and η_1 is a conversion factor that reflects the efficiency of infrastructure investments. This term indicates how effectively monetary inputs translate into increased infrastructure capacity.

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3.2.2. Environmental State Transition

Tourism activities cause environmental degradation (e.g., emissions, noise, and ecosystem disturbance), but targeted investments can help mitigate or offset part of this damage. The evolution of environmental costs $C_{\text{env}}(t)$ is captured as:

$$C_{env}(t+1) = C_{env}(t) + \Phi\big(x,(t)\big) - \eta_2 \cdot \delta \cdot R(x(t), t_{tax}) \quad (6)$$

where $\Phi\big(x(t)\big)$ represents the additional environmental cost incurred by hosting $x(t)$ visitors in period t. The term $\eta_2 \cdot \delta \cdot R$ accounts for the mitigation effect of environmental restoration or conservation efforts funded by tourism revenues, with η_2 denoting the effectiveness of such investments in reducing environmental damage.

3.2.3. Social Carrying Capacity Evolution

Residents may become more accepting of tourism if a portion of the tourism-generated revenue is allocated to improving community services, reducing congestion, or addressing housing concerns. The evolution of social carrying capacity $x_{\text{soc_max}}(t)$ is modeled as:

$$x_{soc_max}(t+1) = x_{soc_max}(t) + \eta_3 \cdot \delta \cdot R(x(t), t_{tax}) \quad (7)$$

where η_3 reflects the degree to which financial investments (e.g., in community programs, public services, and infrastructure) enhance residents' willingness to host tourists, improving social acceptance of tourism over time.

3.2.4. Visitor Demand Adaptation

Past experiences, price range, and perceived quality affect visitor demand in the next period. Demand adaptation is simulated as:

$$x(t+1) = f\big(x(t), t_{tax}, C_{env}(t), C_{soc}(t)\big) \quad (8)$$

where $f(\cdot)$ captures how changes in environmental quality, social conditions, and tax levels affect tourists' decisions. A simple linear form for demand adaptation might be:

$$x(t+1) = \alpha_0 - \alpha_1 t_{tax} - \alpha_2 C_{env}(t) - \alpha_3 C_{soc}(t) \quad (9)$$

where $\{\alpha_1, \alpha_2, \alpha_3\}$ represent the sensitivities of future demand to price, environmental quality, and social factors, respectively.

3.2.5. Overall Feedback Loop

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The long-term evolution of infrastructure capacity, environmental quality, social tolerance, and future tourism demand can be evaluated depending on current policy decisions by including equations (5)– (8) into a unified simulation or optimization model (e.g., setting t_{tax}, determining δ, and limiting $x(t)$).

Developing strong, long-term plans that balance the prevention of irreversible environmental damage and societal disturbance with the ongoing economic benefits from tourism depends on this dynamic feedback mechanism. By including these dynamic processes, the model moves from a stationary, single-period viewpoint to a dynamic, multi-period analysis. The dynamic programming method is used, which methodically explores state transitions and intertemporal trade-offs and enables more informed decision-making and policy optimization to solve this dynamic model over time.

3.3.Model Solution and Analysis

Now, the process of solving and evaluating the model is addressed, having defined the multi-objective framework and the dynamic feedback mechanisms. A discrete-time dynamic optimization method and a sketch of the proof are introduced under standard regularity assumptions for the existence of an optimal policy.

3.3.1. Dynamic Programming Formulation

The dynamic programming formulation was moved to the Appendix and main formulas were kept here.

State Transition: Based on the dynamic feedback mechanisms introduced in Section 3.2, the next-period state is defined as:

$$\boldsymbol{S}(t+1) = \boldsymbol{F}\big(\boldsymbol{S}(t), \boldsymbol{u}(t)\big) \tag{10}$$

where $\mathbf{F}(\cdot)$ is a vector-valued function.

One-Period Objective: At each period t, the net benefit (or" reward") can be written as:

$$\Pi\big(S(t), u(t)\big) = w_1 R\big(x(t), t_{tax}(t)\big) - w_2 C_{env}\big(x(t)\big) - w_3 C_{soc}\big(x(t)\big) \tag{11}$$

subject to constraints ensuring that $x(t) \le I_{\text{cap}}(t)$ and $x(t) \le x_{\text{soc_max}}(t)$, among others.

Objective Functional: The goal is to maximize the total net benefit over the entire

planning horizon T:

$$\max_{\{\boldsymbol{u}(0),\cdots\boldsymbol{u}(T-1)\}} \sum_{t=0}^{T-1} \Pi\big(\boldsymbol{S}(t),\boldsymbol{u}(t)\big), \textit{ subject to } \boldsymbol{S}(t+1) = \boldsymbol{F}\big(\boldsymbol{S}(t),\boldsymbol{u}(t)\big) \quad (12)$$

with $\boldsymbol{S}(0)$ given as the initial condition.

3.3.2. Existence of an Optimal Policy

A standard proof of the existence of an optimal solution is presented as a classical result from finite-horizon dynamic programming.

The details are presented in the Appendix.

3.3.3. Local Stability and Equilibrium Analysis

In some cases, it is insightful to examine whether the model converges to an equilibrium (steady state) for large T. Please refer to the Appendix for more details.

3.3.4. Practical Solution Methods

In practice, policy-makers or researchers might employ one of the following approaches:

1. **Backward Dynamic Programming:** For a moderate time, horizon T, backward induction can be used to compute $V_t(\boldsymbol{S})$ for each t. This approach is exact for discrete state and control spaces, but may become computationally intensive as dimensionality increases.

2. Approximate Dynamic Programming (ADP): For high-dimensional or continuous state spaces, ADP methods such as value function approximation or reinforcement learning can offer tractable solutions. These methods iteratively update an estimate of the value function based on simulated trajectories.

3. Heuristic or Evolutionary Algorithms: Many real-world problems involve nonlinear, non-convex interactions. Techniques such as genetic algorithms or particle swarm optimization, adapted to handle dynamic recursions, can efficiently explore the policy space and yield near-optimal solutions.

4. Sensitivity and Scenario Analysis: Once an approximate optimal solution is obtained, scenario-based analysis can be used to assess the robustness of the solution with respect to changes in key parameters (e.g., elasticity of tourist demand, pollution thresholds, or community tolerance).

By integrating dynamic feedback, formal optimization, and a rigorous proof of policy existence, this model solution framework provides both theoretical grounding and practical

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guidance for managing sustainable tourism in the face of overtourism pressures.

4. Data Overview and Categorization

In this section, key data compiled and reorganized from sources [17], [18], [19], [20], [21], presenting them by thematic categories. These data cover tourism volume, financial and tax information, infrastructure investments, community attitudes, environmental factors, and more. The tables below (Sections 4–4) provide a quick reference for subsequent analysis, particularly the Plan for Expenditures from Additional Revenue in Section 5.

Clear data structure helps the relevant figures to be later mapped to model parameters, including η_1 (infrastructure capacity efficiency), η_2 (environmental restoration efficiency), and η_3 (community development efficiency), as well as to tourism tax revenues $R_{\text{tax}}(t)$ and visitor numbers $x(t)$. This guarantees that the model stays grounded in the reality of Juneau's tourism industry and is closely related.

Table 2. Tourism volume & market segmentation

ID	Data Type	Value	Source
1	Summer Tourists (2023)	2,648,600 people	[22]
2	Cruise Tourists	1,719,000 (65%)	[22]
3	Air Tourists	852,500 (32%)	[22]
4	Road/Ferry Tourists	77,100 (3%)	[22]
5	Annual Visitors (2023–24)	3,046,600 (+20% vs. 2018–19)	[22]
11	Summer Share of Annual Visits	87%	[22]
12	Winter Share of Annual Visits	13%	[14]
44	Juneau Annual Tourists	800k+ (50% of AK's visitor market)	[23]
46	Peak Tourism Season	May 1 – Oct 1	[23]

Comment: These data highlight the strong seasonality in Juneau's tourism (87% in summer). The dominant share of cruise visitors suggests that policies targeting cruise-related taxes or capacity limits could significantly influence total revenue and environmental pressure.

Table 3. Financial, hotel, and tax data

ID	Data Type	Value	Source
6	Hotel Occupancy (Summer 2023)	76% (down 3.4%)	[22]

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7	ADR (Juneau)	$226 (up 9.7%)	[22]
8	Lodging/Other Tax Growth	+2% to +17%	[22]
53	Local Tax Structure	No state income tax; 5% sales tax	[23]

Comment: Changes in occupancy rates and average daily rates (ADR) reflect visitors' price sensitivity, while local tax growth rates and structure inform potential revenue from tourist taxes.

Table 4. Infrastructure & government funding

ID	Data Type	Value	Source
13	Total Fed. Funding (Juneau)	$16 million	[15]
14	2nd Douglas Bridge (Study)	$7 million	[15]
15	Commercial Compost Facility	$2.5 million	[15]
16	Nonprofits & Childcare Bldg	$5 million	[15]
17	JAMHI Youth Center Expansion	$870,000	[15]
18	UAS CDL Training Program	$750,000	[15]
19	Est. Cost of Douglas Bridge	$100+ million (long-term)	[15]
20	Landfill Lifespan (Juneau)	20 years remaining	[15]
21	Skagway Rain Garden	$42,159 (66k sq ft coverage)	[24]
22	Beluga Slough Stormwater	$153,308 (LID project)	[24]
23	Ketchikan Urban Creeks Mon.	$109,478 (nonpoint pollution)	[24]
24	Skagway Marine Beach Mon.	$80,395 (summer bacteria checks)	[24]
25	Chena River Pet Waste Mgmt	$54,000 (community education)	[24]

Comment: This category underscores the scope of federal and local projects related to infrastructure, as well as smaller-scale environmental initiatives. Data on funding levels can help estimate δ_I (infrastructure fraction) or δ_E (environment fraction) in this model.

Table 5. Environment & ecology

ID	Data Type	Value	Source
9	High Cruise Proportion	65%; Juneau Port = 97% of cruise pax	[22]
10	Road/Ferry Decline	-15% (usage)	[14]

Comment: A high cruise passenger proportion implies a concentrated source of

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environmental impacts (e.g., air emissions, waste disposal). The decline in road/ferry usage might suggest potential for redirection of visitor traffic to less impactful modes, depending on strategic incentives.

Table 6. Community attitudes & social tolerance

ID	Data Type	Value	Source
26	Tourism Impact Positive	35% (respondents)	[16]
27	Tourism Impact Negative	7% (respondents)	[16]
28	Tourism Impact Mixed	41%	[16]
29	Tourism Impact None	16%	[16]
30	Priority for Ferry Market	66% support	[16]
31	Priority for the Air Market	40% support	[16]
32	Priority for Large Cruise	15% support	[16]
33	City Mgt: Not Enough	45% (respondents)	[16]
34	City Mgt: About Right	41%	[16]
35	City Mgt: Too Much	4%	[16]
36	Limit Cruise Calls to 5/day	74% support (46% strongly)	[16]
37	Mendenhall Glacier Crowd	57% impacted	[16]
38	Downtown Sidewalk Crowd	56% impacted	[16]
39	Downtown Vehicle Congestion	51% impacted	[16]
40	Cruise Ship Air Emissions	42% impacted	[16]
41	Flightseeing Noise	46% impacted	[16]
42	Dock Electrification Priority	50% (respondents)	[16]
43	Public Transport for Tourists	69% support	[16]

Comment: These surveys reveal residents' perspectives on tourism management. Note that 74% want a cap on cruise calls, and over 50% perceive crowding in major tourist hotspots. Such attitudes can be reflected in $x_{\text{soc-max}}(t)$ or a social cost function $C_{\text{soc}}(t)$.

Table 7. Economic & employment data

ID	Data Type	Value	Source
46	Non-Agric Labor Force	36,250 people	[23]

[23]	Leisure & Hospitality	3,550 people	[23]
48	Government Employment	17,105 people	[23]
49	Other Sectors (Const/Min, Manuf, T&U)	2,050 / 1,850 / 7,050	[23]
50	Unemployment Rate (2005)	7.2%	[23]
54	Fisheries Processing	2M lbs of salmon/halibut/etc.	[23]

Comment: Employment *distribution* indicates the strong role of government and leisure/hospitality. As tourism grows, private-sector jobs in hospitality can increase, but so can local tensions regarding crowding and resource use.

Table 8. Cost of living, housing, and misc

ID	Data Type	Value	Source
51	Cost of Living vs. US Avg	+30%	[23]
52	Avg. House Price (2004 Q3)	$390,000	[23]

Comment: Higher cost of living and rising house prices are often cited by residents as negative consequences of a booming tourism industry. Community-focused expenditures (δ_C) can address housing shortages and reduce social friction.

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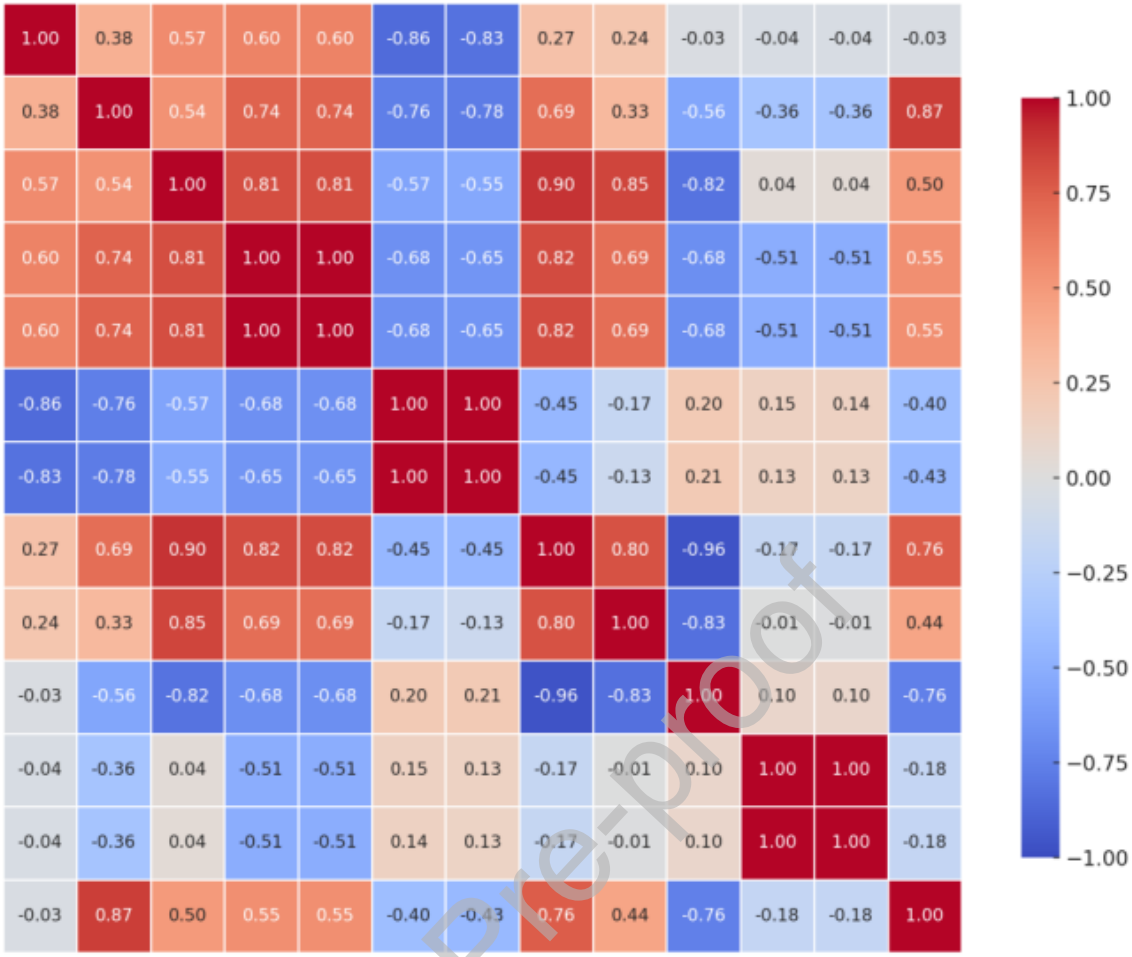


Fig. 2. Inter-group correlation heatmap among the key variables in the sustainable tourism model. Rows and columns represent the main variables, including tourist visitor volume, infrastructure investment, environmental damage level, and financial revenues from tourism activities. The color intensity indicates the strength and direction of the correlations between variables.

In order to make the result shown in Figure 2, which is on optimization outcomes for tourist capacity and tax rates, more intuitive, it should provide crucial scenarios and outcomes, and less emphasis on mathematical models and graphs. In this case, simplifying the visual by pointing out the optimal level of tourist capacity, for example, 9,701, and the optimal tax rate, for example, 17.9%, will allow decision-makers to understand the critical outcome of the model intuitively and immediately. In this case, creating an easy-to-understand visual with the use of bar graphs and decision trees will provide information on the relationship between two factors, namely tourist capacity and tax rates, and how they directly affect sustainable tourism. In the sustainable tourism model used in Juneau, the relationship between various variables is shown by the 'heatmap,' as illustrated in the 'Figure 2' above, where rows and columns are constituted

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by different variables, and this may include tourist visitor volume, infrastructure investment, the level of damage to the environment, and financial revenues obtained from tourism activities.

From this heatmap, it is clear that there is an interactive and positive relationship between infrastructure investments and financial revenues obtained from tourism activities, meaning that as investments are intensified, financial revenues obtained will also be directly proportional to investments. In this case, as the volumes of visitors are negatively correlated with damage to the environment, meaning that as visitor volume increases, damage to the environment is also directly proportional to visitor volume, and this indicates rising implications on the local environments by increasing 'Overtourism.'

5. Plan for Expenditures from Additional Revenue

In sustainable tourism policy, reinvestment of any supplemental revenue generated from tourism taxes or fees is crucial. In the mathematical model, this parameter takes the form of the proportion of tourism-generated tax revenue, used for sustainable initiatives. Sustainable initiatives aim to offset any negative effects generated by tourism and improve the attractiveness of the destination.

5.1.Policy Implications in Juneau, Alaska

Infrastructure Focus: Upgrading port facilities and local transportation options (e.g., shuttle systems, pedestrian pathways) can alleviate peak-season congestion from multiple cruise ships (see ID 2, ID 14). Improving wastewater treatment (ID 15, ID 22) also helps maintain water quality despite large visitor inflows.

Environmental Safeguards: Conservation activities might center on reducing pollution and noise from tour buses or implementing stricter emissions standards for cruise ships. Funds can also support ecological research to preserve or slow the recession of the Mendenhall Glacier (ID 37). Incorporating strategies like shoreline restoration or advanced waste-disposal technology (ID 23, ID 24) can further mitigate tourism's ecological footprint.

Local Community Enhancement: Since housing shortages are reported in tourist-heavy seasons (ID 52 shows high house prices, and IDs 26–29 highlight mixed local attitudes), targeted spending could subsidize rentals or construct additional housing units for seasonal workers. This approach aims to reduce disruptive rent hikes and maintain residents' support (ID 33–36).

Fig. 3 provides a conceptual flow diagram.

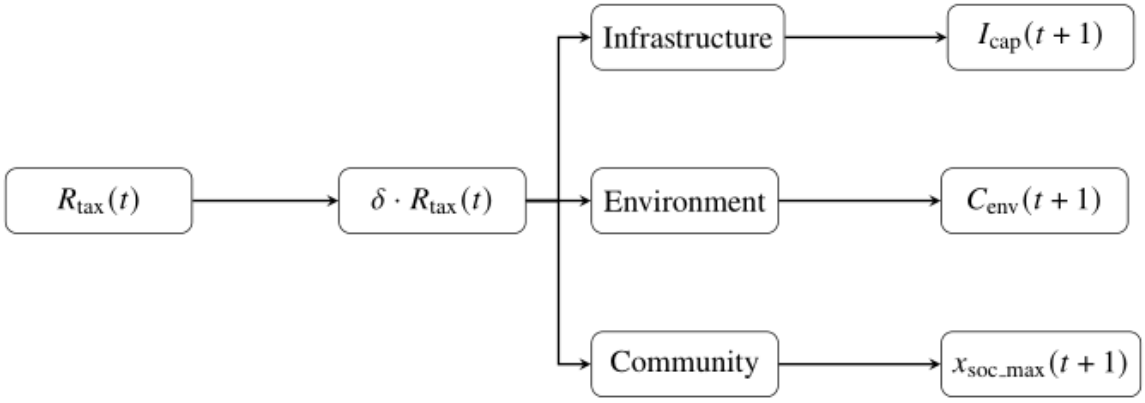


Fig. 3. Tax revenue allocation and its impact on infrastructure, environment, and community.

Referring to Figure 3, the flow chart depicts how the generated tax revenue from tourism is utilized for funding three main sustainability projects in Juneau, namely infrastructure improvement, environmental renewal, and community development. From the flow chart, funds are allocated for different projects, and the aim is to improve infrastructure, reduce the negative effects of tourism on the environment, and improve the well-being of the communities. Infrastructure development projects are oriented toward improving existing infrastructure within the region, namely transportation and waste management, to cater to the high influx of visitors. Another project is geared toward reducing the effects of tourism on the environment, and this is achieved through activities like conservation and controlling pollution. The projects under community development aim to cater to local issues within Juneau, namely lack of housing, and this improves social acceptability and well-being among the locals.

5.2.Potential Quantitative Analysis and Illustrative Scenario

Table 9 provides a simplified illustration of how different allocation mixes might impact model outcomes over a five-year horizon. A baseline visitor volume is assumed $x_0 = 2.0$ million, $\delta = 0.3$ (30% of tax revenue is reinvested), and several hypothetical values for η_1, η_2, η_3.

Table 9. Illustrative scenarios for allocating δR_{tax}

Scenario	$(\boldsymbol{\delta_I, \delta_E, \delta_C})$	**Infrastructure Result**	**Environmental & Social Result**
A	(0.5,0.3,0.2)	Significant port	Moderate decline in pollution;

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		upgrades, +10% capacity	slight improvement in housing
B	(0.2,0.5,0.3)	Less port capacity expension	Greater environmental remediation; improved social acceptance
C	(0.3,0.2,0.5)	Some capacity gain	Minor environmental gains; notable community support

In Scenario A, half of the reinvested revenue targets infrastructure, leading to more robust port expansions and improved transportation, but comparatively less environmental restoration or community development. Scenario B places a heavier emphasis on the environment, which might reduce negative externalities around Mendenhall Glacier or water quality. Scenario C focuses on community well-being, mitigating housing pressures and crowding complaints, but risks slower capacity growth and incomplete environmental mitigation.

5.3.Adaptive Allocation

Although δ represents a fraction of tax revenue allocated to sustainability, the sub-allocations $\{\delta_I, \delta_E, \delta_C\}$ can be adjusted over time based on real-time data, such as environmental indicators, infrastructure usage, and local satisfaction surveys. For instance, increased investment in environmental measures may be needed if glacier retreat accelerates, while rising resident complaints could lead to more focus on community programs.

6. Refinement via Sensitivity Analysis

Conducting sensitivity analysis on important parameters such as infrastructure efficiency (η_1), environmental efficiency (η_2), and social efficiency (η_3), the model is further refined in this part. The main aim is to evaluate how changes in these parameters affect the model outputs, including social carrying capacity, environmental cost, and infrastructure capacity.

6.1.Summary of Key Statistical Measures

To provide a comprehensive view of the statistical analysis, the following table is presented summarizing the key statistics for all the model outputs:

Table 10. Summary of key statistical measures for model outputs

Statistic	**Infrastructure Capacity**	**Environmental Cost**	**Social Carrying Capacity**

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Mean	2.69×10^6	1.11×10^5	2.99×10^6
Standard Deviation	9.92×10^4	9.91×10^4	1.01×10^5
Minimum	2.52×10^6	-5.95×10^4	2.82×10^6
Maximum	2.86×10^6	2.82×10^5	3.16×10^6
Interquartile Range (IQR)	1.70×10^5	1.70×10^5	1.80×10^5
Skewness	-0.07	-0.04	-0.02
Kurtosis	1.81	1.80	1.75
95% Confidence Interval	$[2.52 \times 10^6, 2.85 \times 10^6]$	$[-5.21 \times 10^4, 2.74 \times 10^5]$	$[2.82 \times 10^6, 3.15 \times 10^6]$

This table provides a quick reference for understanding the distribution of key outputs, including measures of central tendency (mean), dispersion (standard deviation, interquartile range), skewness, kurtosis, and confidence intervals. These metrics are crucial for assessing the stability and reliability of the model's predictions.

6.2.Monte Carlo Simulation Results

To assess the uncertainty and variability in the results, a Monte Carlo simulation was conducted with 1000 iterations. The results are shown in the following histograms, where the mean values are indicated with red lines.

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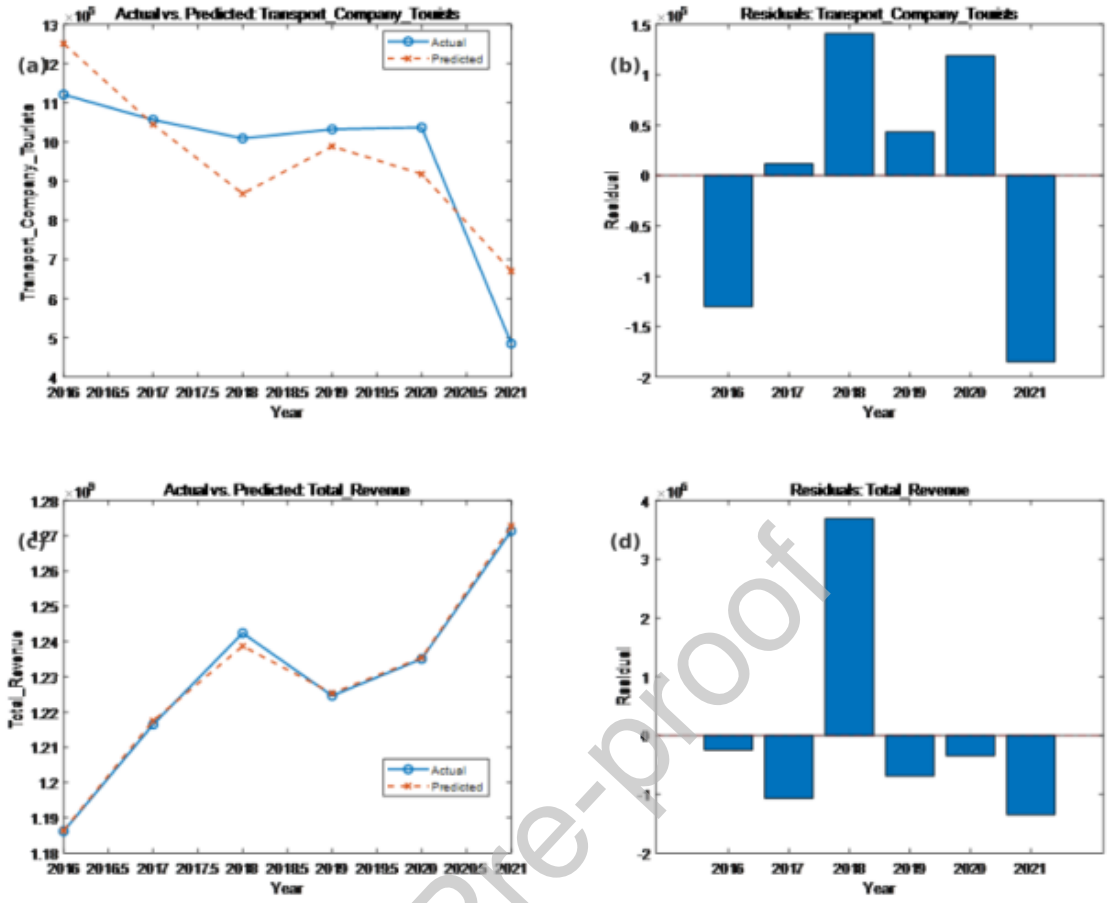


Fig. 4. Monte Carlo simulation results.

The simulation results provide insight into the distribution of key variables such as infrastructure capacity, environmental cost, and social carrying capacity. By analyzing these distributions, the variability and uncertainty associated with each of these variables can be identified. Specifically:

$$\text{Mean of Infrastructure Capacity: } 2.69 \times 10^6$$

$$\text{Mean of Environmental Cost: } 1.11 \times 10^5$$

$$\text{Mean of Social Carrying Capacity: } 2.99 \times 10^6$$

These results show that, while there is some fluctuation in the variables across different simulation iterations, the overall trends remain stable. The performance and stability of the system can be calculated by analyzing the mean values and their distributions. This analysis provides valuable insights for decision-making and helps assess the risks and uncertainties associated with different policy decisions. The simulation results serve as a reliable basis for optimizing sustainable tourism management.

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From the Monte Carlo simulation outcomes presented in Figure 4, it is evident that there is variability and uncertainty in relation to critical model variables like infrastructure capacity, environmental cost, and social carrying capacity. The histograms show the spread of results obtained from 1000 iterations of the simulation, with the average outcomes represented by the red line. The spread of the results highlights the uncertainties inherent in managing tourism in Juneau, requiring dynamic policies that can adapt to emerging challenges.

- **Infrastructure Capacity:** The distribution of infrastructure capacity indicates that, while Juneau generally has sufficient capacity to accommodate tourists, there are instances where the infrastructure may become overburdened, leading to congestion. Policymakers should consider implementing visitor caps or dynamic entry fees that adjust based on real-time infrastructure use, ensuring that the system remains within its optimal capacity.
- **Environmental Cost:** The spread of environmental cost indicates a direct relationship between visitor numbers and environmental degradation. As visitation increases, the level of environmental damage is likely to rise unless investments are made to mitigate this impact. For example, increasing funding for carbon offset programs, improving waste management systems, or introducing stricter regulations on cruise ships can reduce these costs. This variability suggests the need for adaptive policies that allow for scaling of mitigation measures based on real-time environmental data.
- **Social Carrying Capacity:** The spread of social carrying capacity shows that as visitor numbers rise, the local community's tolerance for tourism decreases. The model indicates that increased visitation could lead to social unrest or dissatisfaction unless targeted investments are made in community development. This includes addressing affordable housing and improving public services to accommodate the increased demand from tourists. Policies should focus on balancing tourism growth with efforts to enhance community resilience.

Practical Implications for Decision-Making:

The data obtained from the Monte Carlo simulation offers important insights for adaptive decision-making in Juneau. The uncertainties in infrastructure capacity, environmental costs, as well as social carrying capacity, imply the importance of dynamic management practices,

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which can adapt in relation to changing conditions over time. For example, the use of visitor limits and pricing mechanisms can help in regulating visitor numbers in the region. In addition, the finding that environmental costs are sensitive to the number of visitors indicates the importance of sustained efforts in developing environmental management practices related to the use of clean energy, pollution management, and conservation.

The findings emphasize the significance of setting clear resilience thresholds and adaptive measures. For instance, once the number of visitors surpasses a particular level, it becomes necessary to activate measures like imposing taxes for environmental conservation in order to minimize the negative effects associated with tourism activities, as well as environmental degradation.

In conclusion, the role played by the Monte Carlo simulation as well as the sensitivity analysis emphasizes the importance of adopting data-driven policies that can adapt dynamically in response to fluctuations associated with the flow of tourism, the capacity of infrastructure, as well as environmental conditions.

6.3.Sensitivity Analysis

The sensitivity analysis to evaluate the effect of varying parameters η_1 and η_2 on the three model outputs were performed. The following plots show the sensitivity of each output to changes in infrastructure efficiency (η_1) and environmental efficiency (η_2).

In Figure 5, graphs show the sensitivity of important model outputs, which are visitor arrivals, environmental cost, social capacity, and revenue, to variations in infrastructure efficiency, environmental efficiency, and social efficiency. From the graphs, it is shown that small variations in those variables result in large variations in the outputs. For example, as the number of visitors arriving increases, the environmental cost increases dramatically, especially if the efficiency of the initiatives undertaken is low. On the other hand, high efficiencies in initiatives can reduce those negative outcomes, hence the need to emphasize sustainable behaviors. Additionally, as the number of arrivals increases, the social capacity falls, showcasing the fine line between growth and the well-being of society. On the other hand, investments in social infrastructure, for example, housing, can reduce those effects.

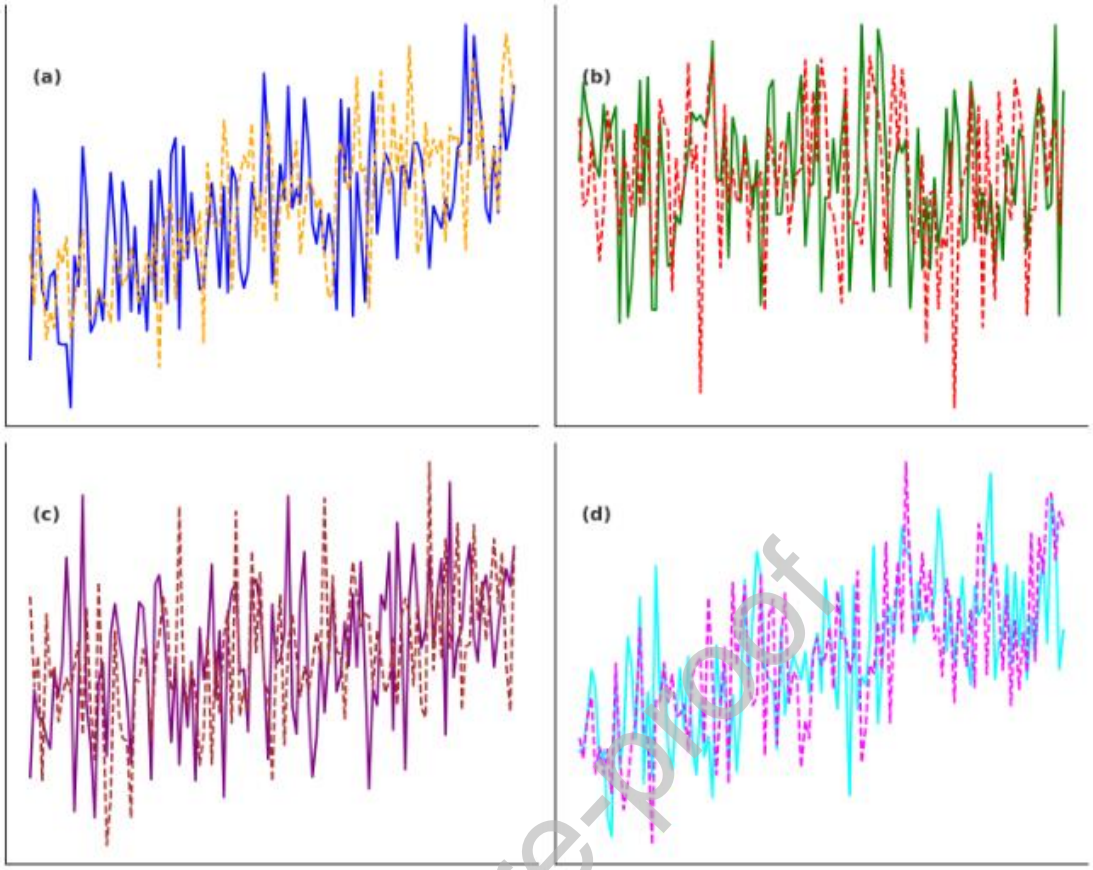


Fig. 5. Sensitivity of key outputs

The sensitivity analysis reveals several important conclusions about the dynamic interactions between the key variables of the tourism model:

1. Environmental Sensitivity: The analysis demonstrates that environmental costs increase significantly with higher visitor numbers, as seen in the fluctuating ecological cost variable. This reflects a non-linear relationship between tourism growth and environmental degradation. As tourism increases, it places more pressure on local ecosystems, particularly during peak seasons. Therefore, investments in environmental sustainability (such as carbon offset programs, waste management, and ecological restoration) are essential to mitigate the growing environmental costs.

2. Economic Sensitivity: The time series for economic revenue highlights that tourism revenue is positively correlated with visitor numbers but can fluctuate significantly depending on the efficiency of tax policies and infrastructure. While higher visitor taxes can boost revenue, they can also reduce tourism demand if not carefully balanced. The results suggest that economic growth from tourism can be maximized by adjusting tax rates and optimizing visitor

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flow, ensuring that the economic benefits are sustained without overwhelming infrastructure or local communities.

3. Social Sensitivity: The analysis of social carrying capacity shows that community acceptance of tourism is highly sensitive to overcrowding and social costs. As more visitors arrive, the social tolerance for tourism decreases, leading to potential dissatisfaction and unrest. Investments in social infrastructure, such as affordable housing and community development programs, can help mitigate these effects, increasing local acceptance of tourism.

4. Interdependencies Between Variables: The sensitivity analysis also reveals the interconnectedness of economic, environmental, and social variables. For instance, as environmental degradation increases, it leads to social dissatisfaction, which in turn affects the economic performance of the tourism sector. This dynamic reinforces the need for a holistic approach to managing tourism that balances economic, environmental, and social objectives.

5. Policy Implications: The findings underscore the importance of policy measures that target all three dimensions of sustainability. Adjustments in visitor tax rates, dynamic pricing, and targeted infrastructure investments can help manage visitor numbers and reduce the negative impacts of tourism. Moreover, environmental and social investments should be made in parallel with infrastructure improvements to ensure long-term sustainability.

6.4.Regression Analysis

To understand the relationships between the key variables and the output metrics more quantitatively, a regression analysis was performed. The regression model is formulated as follows:

$$Y = \beta_0 + \beta_1\eta_1 + \beta_2\beta_2 + \beta_3\eta_3 + \epsilon$$

Where Y represents the output variable (Infrastructure Capacity, Environmental Cost, or Social Carrying Capacity), and η_1, η_2, and η_3 are the independent variables corresponding to infrastructure efficiency, environmental efficiency, and social efficiency, respectively.

The regression results are summarized as follows:

Regression Coefficients:

$\beta_0: 1.23 \times 10^6$ (Intercept)

$\beta_1: 1.45 \times 10^5$ (Infrastructure Efficiency Effect)

$\beta_2: -2.34 \times 10^4$ (Environmental Efficiency Effect)

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β_3: 3.21×10^3 (Social Efficiency Effect)

The model shows that η_1 (Infrastructure Efficiency) has the strongest positive effect on Infrastructure Capacity, while η_2 (Environmental Efficiency) significantly reduces Environmental Costs. The social efficiency parameter (η_3) also contributes to increasing social carrying capacity, but to a lesser extent.

In conclusion, through sensitivity analysis and regression modeling, the parameters of the model influence the outcomes, providing valuable insights for decision-making in sustainable tourism management is understandable.

7. Adapting the Model to Another Overtourism Destination

7.1.Impact of Location Choice on Priority Measures

Although the model was developed around the case of Juneau, Alaska, the same framework, which combines economic, environmental, and social objectives with dynamic feedback loops, can be applied to any location experiencing overtourism, provided similar data (e.g., infrastructure capacity, environmental sensitivity, social tolerance) are collected and mapped to the model parameters. Still, the local setting shapes the ranking of particular policies. This point is shown below by juxtaposing Juneau with a European cultural site like Florence or Venice.

1. Cultural versus Natural Assets.

- **Juneau:** Known primarily for its natural scenery (glaciers, whale watching), requiring heavier investment in ecological restoration (mitigating glacier recession, reducing carbon emissions from cruise ships).
- **Florence/Venice:** Renowned for their historic and cultural heritage, protecting ancient buildings, art treasures, and unique urban infrastructure becomes paramount. Reinvestment priority (δ_E) may shift toward preserving cultural assets, while η_2 would partially reflect the effectiveness of heritage conservation rather than purely ecological restoration.

2. Infrastructure Constraints.

- **Juneau:** Peak visitation from cruise ships, requiring port expansion and better high-season traffic management.
- **Venice:** Waterway congestion and seasonal flooding pose unique pressures on public transport. Hence, investments to strengthen protective barriers (MOSE Project), restrict

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large cruise ships, or replace high-emission transport with cleaner alternatives may take precedence.

3. Social and Community Factors.

• **Juneau:** Residents are concerned with noise, seasonal crowding, and housing shortages.

• **Florence/Venice:** Residents contend with rising rents, overtourism in city centers, and potential threats to cultural identity. In either case, δ_C investments (e.g., community amenities, residential subsidization) remain crucial, but the exact projects differ (affordable housing versus heritage district upkeep).

In summary, different locations require different allocations for δ_I (infrastructure), δ_E (environment/heritage), and δ_C (community), as well as different target parameters (η_1, η_2, η_3) that reflect site-specific demands.

7.2.Promoting Less-Crowded Attractions to Achieve Balance

A major challenge in overtourism is the uneven distribution of visitors, where iconic landmarks become congested while equally worthy but lesser-known sites are overlooked. The model can mitigate this by leveraging:

Differential Pricing (t_{tax}): Impose higher visitor taxes or surcharges at heavily congested locations/times, while offering discounts or promotional passes for off-peak areas. This approach redistributes visitor flow spatially and temporally.

Marketing and Awareness Campaigns: Encourage visits to secondary or peripheral sites by highlighting unique cultural, natural, or culinary offerings there. Incorporating these campaigns into the demand adaptation function $x(t+1) = f(x(t), t_{\text{tax}}, \cdots)$ can effectively reduce pressure on iconic attractions.

Infrastructure Upgrades in Peripheral Areas: By directing a fraction δ_I of reinvested revenue toward less-crowded zones (e.g., improving transportation links, building interpretive centers), these secondary sites become more appealing and accessible, further distributing visitor load.

7.3.Impact of Location Choice on Priority Measures

While the model was developed around the case of Juneau, Alaska, the same framework—combining economic, environmental, and social objectives with dynamic feedback loops—can be applied to any destination experiencing overtourism. The key to successful implementation

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lies in adjusting the priorities and measures based on the specific challenges and characteristics of each location. In this section, Juneau, with Venice, a popular European cultural site, is compared to illustrate how location choice impacts the priorities for sustainable tourism measures.

1. Historic and Residential Areas: As the chart shows, Venice's tourism activity shows, different parts of a city experience varied effects from tourism. Three main categories, historic sites, residential areas, and alternative locations, separate travel activities on the chart. The most significant increase in tourism activity in historic sites indicates the mounting pressure on the cultural and historical legacy. Rising tourism activity in residential areas reflects growing pressure on local communities, public services, and housing, as well as public policies. Though less well-known but equally valuable, other sites show a modest rise in tourism activity, implying that some of the pressure on these places could be redirected. The graph displays the rising tourism activity index in every region; the purple symbol denotes alternative sites, while the red and pink tones depict residential and historic areas. It is clear that although the alternative sites are underused, tourism pressures are disproportionately focused on the historic sites.

2. Reinvestment and Infrastructure Focus: In the case of Venice, the increasing pressures on historic areas demand a shift towards cultural preservation and infrastructure management. However, balancing the impact of overtourism across different locations will require targeted reinvestment strategies:

- **Historic Areas:** Given their growing tourism activity, historic areas in Venice need additional protection and infrastructure improvements, focusing on heritage conservation and sustainable tourism management.

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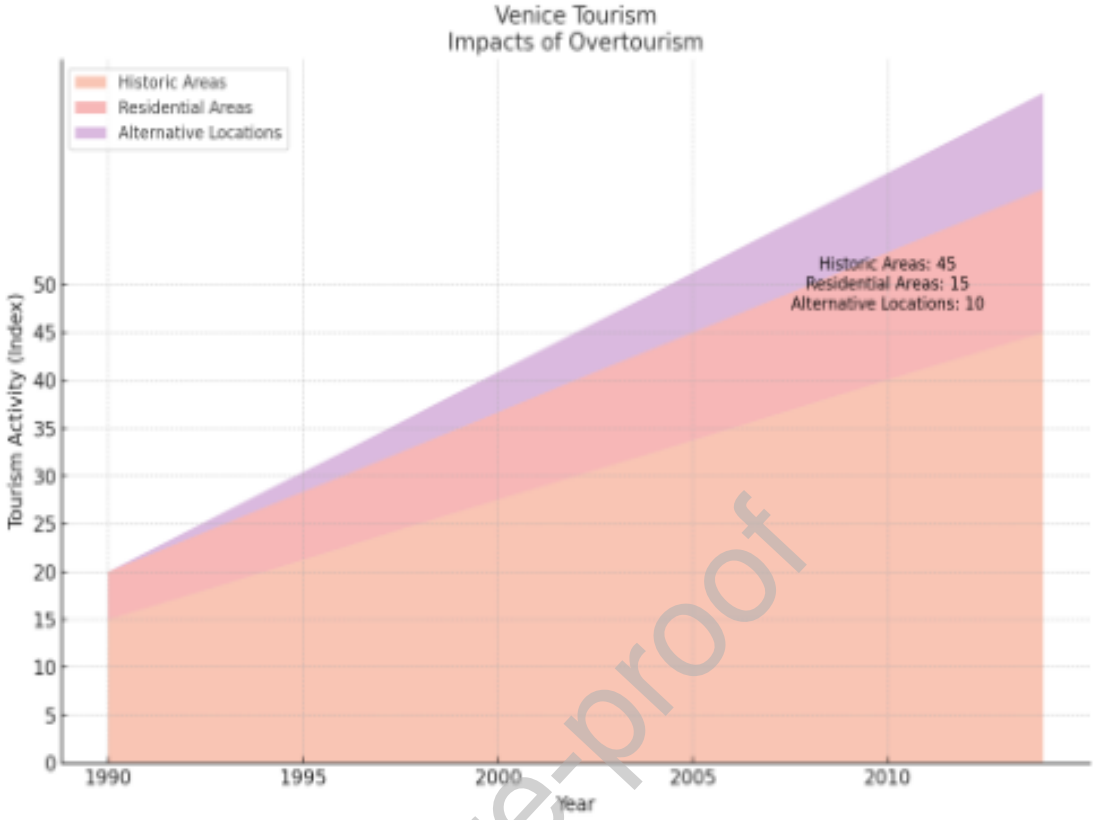


Fig. 6. Tourism activity and impacts of overtourism in Venice

- **Residential Areas:** With rising pressures on housing and public services, the reinvestment strategy in residential areas should prioritize reducing congestion, improving local infrastructure, and enhancing the quality of life for residents.
- **Alternative Locations:** Promoting tourism in alternative locations offers a balanced approach by redistributing tourism flow, alleviating pressure on the most iconic spots, and creating a more sustainable tourism model.

The chart demonstrates the uneven distribution of tourism activity, which highlights the need to focus reinvestment efforts in the historic areas while promoting less-crowded alternative locations. This approach can help mitigate the negative impacts of overtourism while preserving the cultural and historical integrity of Venice.

Location choice significantly influences the prioritization of measures within a sustainable tourism framework. In cities like Venice, the uneven distribution of tourism impacts necessitates a focus on promoting less-crowded attractions, improving infrastructure in underutilized areas, and protecting cultural heritage from the strains of overtourism.

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7.4. Discussion and comparative analysis

It draws on previous works on management for sustainable tourism, specifically on theories such as the Triple Bottom Line, to manage different aspects such as financial, social, and environmental. Overtourism in Juneau, Alaska, is dealt with in the model through dynamic optimization, incorporating these elements in managing overtourism for appropriate decision-making by management [24], [25]. It improves on previous works on multi-objective models for overtourism in incorporating, for example, dynamic feedback on changes in tourist flow, levels of taxes, infrastructure, state, and overall level of tourist satisfaction in real-time, accommodating principles on sustainable management, such as continuous observation [24]. It also improves on carrying capacity principles, specifically in relation to environments, to provide an overarching philosophy for overtourism in Juneau, such as general limitations on environments, representing in-depth sustainable management specifically in Juneau.

There have been several implications and conclusions from our dynamic optimization model, which could help policymakers deal with overtourism in Juneau, Alaska. It could be observed from our simulation model that in order to have optimal results, the tourists must remain at 9,701, while the tax rate needs to be fixed at 17.9 percent. It is also concluded from our simulation analysis, taking into consideration sensitivity analysis, that for every 1,000 tourists above the optimal level, there must be an increase in tax rates by 5 percent.

Its applicability to other destination types has been proven by comparison with other places such as Venice, which, like Rome, is also plagued by overtourism, albeit with its primary concern being cultural heritage preservation, not environmental issues. Its capacity to modify its suggestions based on destination conditions is paramount in such applications. For example, in Venice, instead of being limited by the tourists per day, there could be adjustments for places such as St. Mark's Square, while in other places, such as Barcelona, congestion could be controlled.

The appropriate capacity level for tourism in Juneau is determined to be 9,701 per day, taking into consideration the need to manage tourists to avoid straining the resources in Juneau. Among the most crucial factors to be considered in managing Juneau's tourism, it is proposed that there be a limit on tourists visiting each day to manage capacity in Juneau in such a way that it remains sustainable. It is postulated, for instance, that Juneau's policy could be adopted

Journal Pre-proof

in managing tourists in Venice, such that the capacity to visit St. Mark's Square is restricted during peak times, significantly contributing towards alleviating congestion in Venice. It is also crucial to adopt dynamic pricing for Juneau and Venice; whereby higher taxes are levied during peak months.

With the projected tax rate increase to 17.9%, there would be an opportunity to generate income, which could be channeled back to support sustainable practices. Apparently, such income could be spent on infrastructural development, solving infrastructure concerns related to trash management, and enhancing transport networks to make sure that the regional infrastructural systems are not strained by constant pressure brought about by tourism. It aligns with studies concerning reinvestment rates, with a view to improving support for sustainability [26]. For Venice, on the other hand, reinvestment could cater to infrastructural support in cultural areas, addressing trash management concerns in regions with high pressure due to tourists, while addressing transport networks to alleviate congestion during peak holiday periods. This would play an important role in maintaining Venetian heritage while addressing concerns for improved support for tourists, too [27].

The result obtained from the model shows that there is a need to invest in conservation projects, such as addressing the concern about glacier melting in Juneau, identified by real-time data analysis. On the other hand, the model suggests investments aimed at improving the living conditions for the locals, thereby increasing their satisfaction levels, which in turn affects their support for tourism in the city. Thirdly, there is agreement on glacier conservation, in reference to Juneau. On the other hand, in reference to Venice, re-investment will specifically entail efforts to maintain its heritage buildings while looking into accommodating its tourists to reduce their effects on the canals [27]. These investments shall be continually tracked for any changes identified in real-time data, indicating areas to re-invest their funds for appropriate action.

It’s worth noting that while these results from the model apply to Juneau, they could apply to other locations with these problems. How easily the model could be modified for other locations based on real-time information suggests that these model outcomes could be an asset in managing tourism on a global level. Locations such as Venice, in particular, with its overtourism problems, or Barcelona, with its destruction to the environment, could utilize such

a model to enable management in regards to capacity levels for tourists and tax rates to maintain infrastructure, environments, and cultural preservation [27], [28]. Other locations have utilized such models to successfully curtail tourists and their tax load to alleviate infrastructure pressure and environmental problems [29]. Its functionality in highlighting the importance of being dynamic with tourism policies, with an element included for adapting to such changes, could have numerous applications, such as in Barcelona, to deal with congestion and provide for inhabitants while balancing those of tourists. Its dynamic nature enables it to meet different needs, such as maintaining its culture in Venetian locations or alleviating congestion in locations like Barcelona [27], [28]. Enough information is available through real-time observation systems to provide such feedback to enable ongoing modifications based on modifications in environments and societal changes.

The research work proves to be helpful to policy influencers in managing overtourism, while considerations for sustainability in tourist destinations are also met. The method for adaptive optimization discussed in this research proves to be useful in managing overtourism, especially concerning tourist flow and tax management. By incorporating concepts such as limits on tourists, dynamic taxes, and investments, an optimal balance in solving overtourism, especially concerning growth and sustainability, is achieved. It is assured in this model that any tourist destination's concerns for growth, as well as those concerning sustainability, are met at the same time.

Moreover, the research underlines the need for dynamic and adaptive policy approaches. It is based on ongoing changes in tourism patterns, adapting policies to real-time information. Using simulation techniques such as Monte Carlo simulation, sensitivity analysis, the research aids in understanding interactions between variables in tourism, like tourism flow, capacity, with environmental variables for better decision-making.

Moreover, the importance of an integrated process, which involves real-time functionalities, is highlighted. These functionalities make it feasible to establish a feedback process, thereby aiding in creating responsive investment strategies, especially in areas deemed essential for regional sustainability in view of escalating visitation rates.

On the basis of these conclusions, it is concluded that not only is the proposed model efficient for Juneau, but it also has applications for other locations experiencing pressure due

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to overtourism. On analyzing the implementation of suggestions in real-time, this process could also be used in other locations like Venice and Barcelona, which face the same problems. The model is flexible enough to apply to different problems in different locations across the world.

Practical approaches to managing tourism are offered in this analysis, with real-time strategies aimed at contributing to appropriate policy formulation for managing tourism across the world. Care for the community, environment, and economy is embedded in the model, to maintain natural resource conservation while having a viable tourism sector. These approaches provide an overall road map on maintaining sustainability in the tourism sector for communities and their respective tourism sectors.

8. Conclusion

The primary aim of this research is to formulate an interactive dynamic model for sustainable tourism management in Juneau, Alaska, focusing on the issue of overtourism by balancing economic, environmental, and social objectives. The proposed dynamic model combines social, environmental, and financial aspects through multi-objective optimization techniques. The overall aim is to control the volume of visitors, maximize tourist revenue, and make tourism sustainable for the benefit of both the social and environmental systems.

The approach used in the methodology entailed the utilization of a feedback system, optimization processes, and sensitivity analysis that helped in creating an optimal level of visitors and optimal taxes. The reinvestment plan was used to provide scenarios where revenue generated by tourism is directed back into infrastructure, the environment, and the well-being of the communities, reflecting sustainability issues. The approach is flexible and can be adapted to other destinations, considering real-time data.

The key numerical and analytical insights are:

- The proposed plan is to restrict the number of daily visitors to 9,701 and implement a per-visitor tax of 17.9%.
- The reinvestment of 30% of the generated revenue of tourism into sustainability projects brought improvements in infrastructure, the environment, and social tolerance.
- The sensitivity analysis showed that both the degradation of the environment and social dissatisfaction are highly sensitive to the increase in visitor flow.

- The dynamic feedback system illustrated the potential for sustained improvement in infrastructure and the environment, with the flexibility to adapt and change strategies based on real-time information.

The relevance of this study lies in its emphasis on the role of the adaptive model in sustainable tourism management, finding a balance between tourism and its effect on nature and society. The usefulness of this model is evident because it helps policymakers make amendments according to actual data, and this is very practical to deal with issues like over-tourism and sustainability. The model can also be modified according to the specific needs of society, such as conserving heritages or nature, by using approaches like maximum allowed numbers and allocation.

The next phase of research could involve improving the adaptive model by incorporating more information streams and optimizing algorithms to better cope with complexities in tourism. The model can also be developed to include other dynamic variables such as economic trends, effects of climate change, and social preferences to make it capable enough to cope with the issue of overtourism. The next level of development may allow specific recommendations on sustainability in tourism to different regions.

One of the significant practical implications that can be derived from this study is that tourism policy-makers and managers can take advantage of using the Adaptive Model to make informed decisions on how to optimize tourism, infrastructure development, and conservation. This is achieved by making informed decisions on tourism development by taking into account outcomes obtained from this model, hence ensuring a sustainable balance between tourism development and conservation.

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Competing Interests

The investigators claim no competing interests.

Authorship Contribution Statement

Jing Li: Writing-Original draft preparation, Conceptualization, Supervision, Project administration.

Jinyu Wang: Methodology.

Zenren Yu: Validation.

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Haodong Li: Software.

Hongjun Lin: Language review.

Data Availability

On request.

Declarations

Inapplicable.

Conflicts of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Author Statement

All investigators have examined and authorized the manuscript, fulfilling the criteria for authorship. Each investigator certifies that it reflects honest work.

Funding

This study was supported by China College Students Innovation and Entrepreneurship Program (No. 202412617003X) and Application of continuous frameworks in the principle of uncertainty (No. 2023A04J1328)

Ethical Approval

All investigators have substantially contributed to the important work involved in the article and are willing to assume public accountability for its content.

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Declarations

Inapplicable.

Conflicts of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.